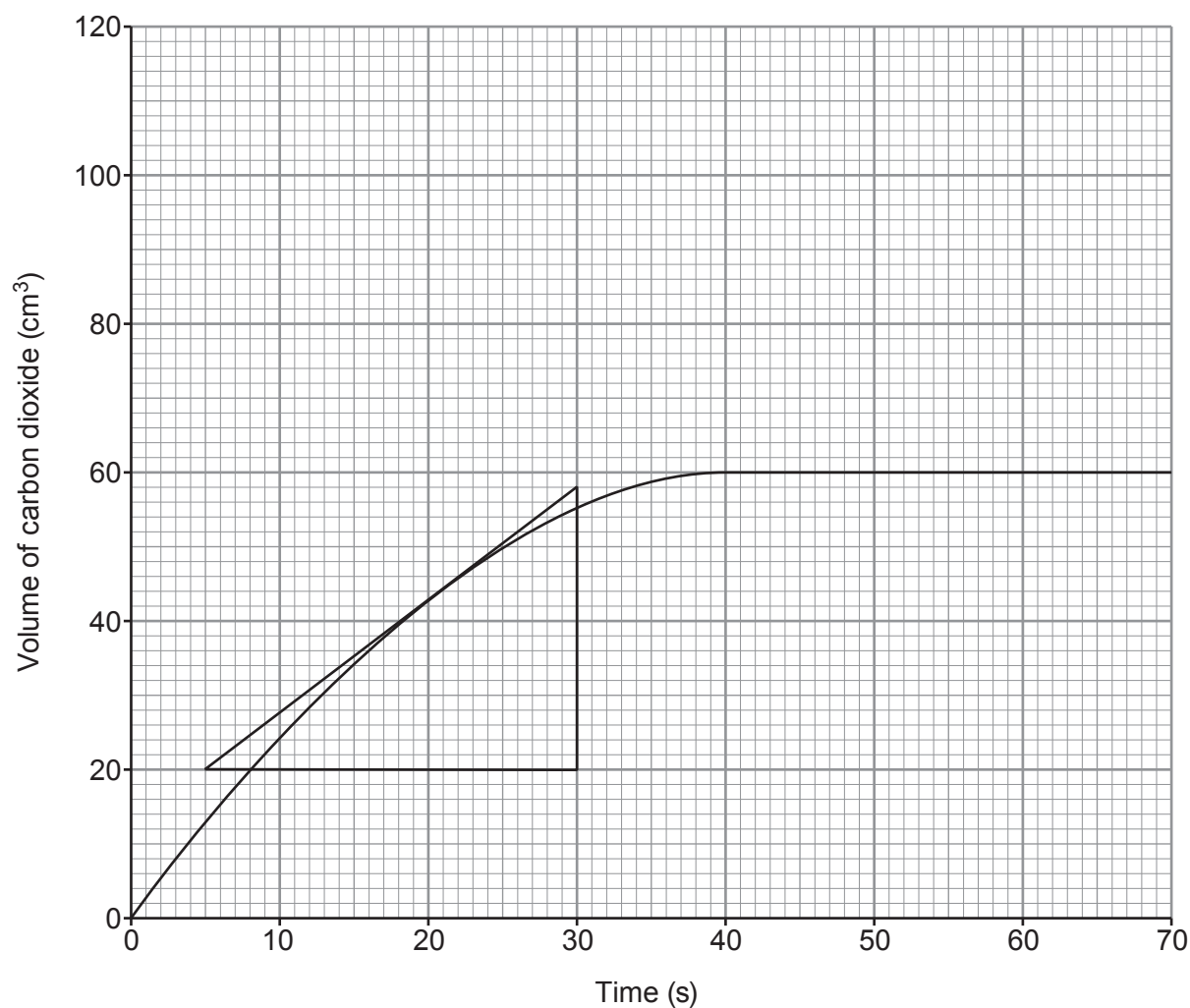


11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol/dm³ hydrochloric acid.



- (a) Use the tangent drawn to calculate the rate of the reaction at 20 s.

[2]

Rate = cm³/s



- (b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]

- (ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

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END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1 H Hydrogen 1																			
										4 He Helium 2									

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number